

## Detailed Design Procedure LT3650

From datasheet:

$$C_{in} := 10 \mu F \quad C_{Bat} := 10 \mu F \quad C_{Bat2} := 100 \mu F$$

Design inputs:

$$I_{max} := 2 A \quad \Delta I_{max} := 0.25 \cdot I_{max} = 0.5 A \quad V_{Bat} := 8.2 V$$

$$V_{in} := 12 V \quad V_f := 420 mV$$

Calculated values:

$$R_{sense} := \frac{0.1 V}{I_{max}} = 0.05 \Omega$$

$$L := \left( \frac{10}{\Delta I_{max}} \right) \cdot R_{sense} \cdot (V_{Bat} + V_f) \cdot \left( 1 - \frac{V_{Bat} + V_f}{V_{in} + V_f} \right) \cdot \frac{\mu H}{\Omega^2} = 2.637 \mu H$$

Calculated value is too small for a robust system, therefore the proposed value from datasheet will be used.

$$L := 10 \mu H$$

$$I_{diode} > \frac{I_{max} \cdot (V_{in} - 5.65 V)}{V_{in}} \rightarrow I_{diode} > 1.0583333333333333 \cdot A$$

It means  $I_{diode} > 1.1 A$